

Unit 5

Human-Centered Aerospace Design

Use user research, ethics, sustainability, project planning, testing, and presentation skills to complete a meaningful aerospace design challenge.

Student goal

Design with people, evidence, and impact in mind.

Capstone focus

From need finding to portfolio reflection.



Unit 5 Learning Arc

This unit turns your engineering skills into a complete human-centered capstone story.

1

Understand people

Stakeholders, users, interviews, and problem statements

2

Define requirements

Criteria, constraints, ethics, risk, sustainability, and success metrics

3

Plan and build

Team norms, project management, concepts, CAD, and prototypes

4

Test and communicate

Data, iteration, final presentation, showcase, and portfolio



By the end of Unit 5, you will be able to explain not just what you made, but why it matters and how evidence shaped it.

LESSON 5.1

Human-Centered Aerospace Design

Essential question

How do aerospace designers make solutions that work for real people, not just technical requirements?

What you will create

Human-centered design mindset notes



What you need to understand

How do aerospace designers make solutions that work for real people, not just technical requirements?

Big ideas

- Human-centered design starts with people, context, and purpose.
- Aerospace problems often involve safety, reliability, access, cost, and communication.
- A successful solution balances user needs with technical feasibility.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Look beyond the object

Identify who uses, supports, maintains, pays for, or is affected by the design.

2 Separate assumptions from evidence

Do not treat what you think users need as proof. Gather evidence.

3 Translate needs into requirements

Turn user problems into design criteria that can guide decisions.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Human-centered design mindset notes

Evidence checklist

- Define human-centered design in your own words.
- Identify at least three user groups in an aerospace system.
- Explain how one design choice affects a user.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.2

Stakeholders, Users & Clients

Essential question

Who is affected by an aerospace design problem, and how can their needs conflict?

What you will create

Stakeholder map for a client problem



What you need to understand

Who is affected by an aerospace design problem, and how can their needs conflict?

Big ideas

- A user directly interacts with the design; a stakeholder is affected by it.
- A client defines the problem but may not be the only important voice.
- Needs may conflict, so engineers must prioritize and justify choices.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Map the people

List direct users, indirect users, operators, maintainers, decision makers, and people affected by failure.

2 Name the need

Write needs as what people must accomplish, not as a specific product.

3 Watch for conflicts

A feature that helps one stakeholder may create cost, safety, or usability issues for another.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Stakeholder map for a client problem

Evidence checklist

- Create a stakeholder map.
- Label users, stakeholders, and client.
- List at least two conflicting needs.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.3

User Research & Empathy Interviews

Essential question

How can you learn what users actually need before designing a solution?

What you will create

Interview or observation plan



What you need to understand

How can you learn what users actually need before designing a solution?

Big ideas

- User research is evidence gathering, not opinion collecting.
- Open-ended questions reveal context, frustrations, and priorities.
- Observations can show problems users may not mention in an interview.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Ask open questions

Use prompts such as “Tell me about...” and “What makes this difficult?”

2 Avoid leading questions

Do not ask questions that push users toward your preferred solution.

3 Capture evidence

Record quotes, actions, pain points, and repeated patterns.

Engineering habit



Evidence loops back into the next decision.

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What your work should show

Use this lesson to produce: Interview or observation plan

Evidence checklist

- Write 6–8 interview questions.
- Identify what evidence each question should reveal.
- Plan how notes will be documented.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.4

Problem Statements & Design Briefs

Essential question

How do you turn user research into a clear engineering problem?

What you will create

Draft design brief



What you need to understand

How do you turn user research into a clear engineering problem?

Big ideas

- A strong problem statement names the user, need, and context.
- A design brief explains what the solution must accomplish without naming the final design too early.
- Clear problem framing prevents teams from solving the wrong problem.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Use a problem frame

User + need + context + reason the problem matters.

2 Avoid solution language

Do not write “build a bracket” if the real need is alignment, safety, or repeatability.

3 Check the scope

The problem must be large enough to matter and small enough to solve.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Draft design brief

Evidence checklist

- Write a first draft problem statement.
- List the client or stakeholder need.
- Identify what success would look like.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.5

Criteria, Constraints & Success Metrics

Essential question

How can design requirements be written so a solution can be tested fairly?

What you will create

Requirements table

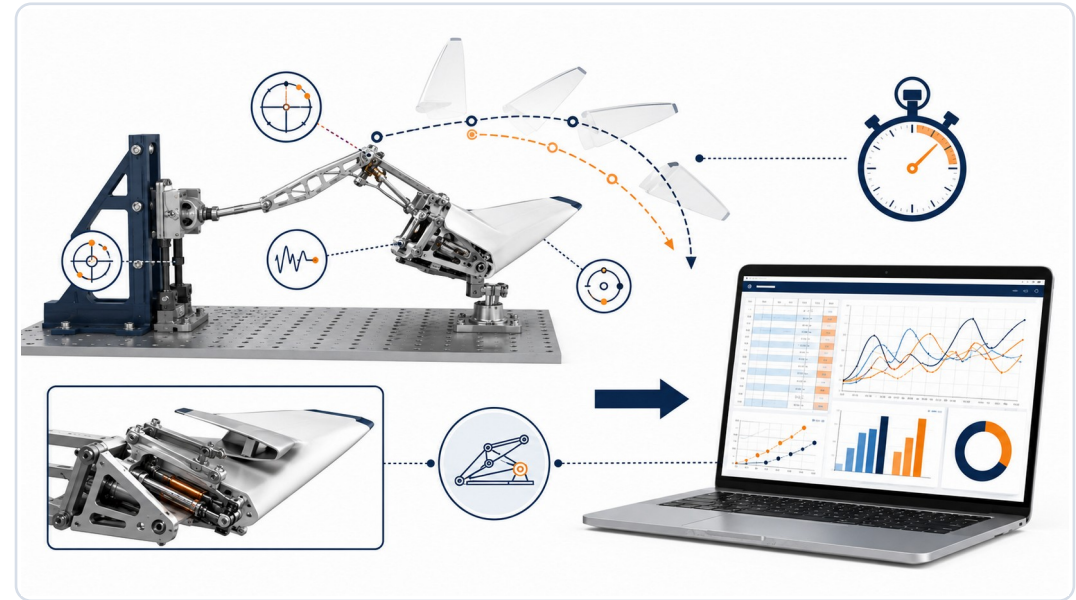


What you need to understand

How can design requirements be written so a solution can be tested fairly?

Big ideas

- Criteria define what a successful solution should do.
- Constraints define limits such as time, material, cost, tools, or safety.
- Success metrics make requirements measurable.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Make criteria measurable

Replace “stable” with a measurable test such as “does not tip during three trials.”

2 Separate must-have from nice-to-have

Teams need to know what is required versus preferred.

3 Connect metrics to tests

Every major metric should have a planned way to measure it.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Requirements table

Evidence checklist

- Create a requirements table.
- Label criteria and constraints.
- Write at least three measurable success metrics.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.6

Engineering Ethics in Aerospace

Essential question

What responsibilities do engineers have when designs affect safety, trust, and public systems?

What you will create

Ethics scenario response



What you need to understand

What responsibilities do engineers have when designs affect safety, trust, and public systems?

Big ideas

- Engineering decisions can affect safety, cost, access, and public trust.
- Ethical design requires honesty about limitations, risks, and evidence.
- Aerospace systems require careful documentation because failures can have serious consequences.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Identify who could be harmed

Consider users, operators, the public, and the environment.

2 Use evidence honestly

Do not exaggerate test results or hide known failure modes.

3 Escalate concerns

When a design may be unsafe, engineers must communicate clearly and early.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Ethics scenario response

Evidence checklist

- Analyze one aerospace ethics scenario.
- Identify affected stakeholders.
- Write a justified decision or recommendation.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.7

Sustainability & Life-Cycle Thinking

Essential question

How can engineers reduce waste and environmental impact across a product's life?

What you will create

Life-cycle analysis sketch



What you need to understand

How can engineers reduce waste and environmental impact across a product's life?

Big ideas

- A product has impacts before, during, and after use.
- Material choice, manufacturing method, repairability, and disposal all matter.
- Sustainable design balances performance with responsible resource use.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Trace the life cycle

Raw materials → manufacturing → use → maintenance → end of life.

2 Look for waste points

Identify where material, energy, time, or replacements are wasted.

3 Design for repair or reuse

A part that can be replaced or repaired may reduce total impact.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Life-cycle analysis sketch

Evidence checklist

- Create a life-cycle diagram.
- Identify one design choice that reduces waste.
- Explain a tradeoff between performance and sustainability.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.8

Safety, Risk & Failure Modes

Essential question

How can you predict what might go wrong before testing a prototype?

What you will create

Failure mode table

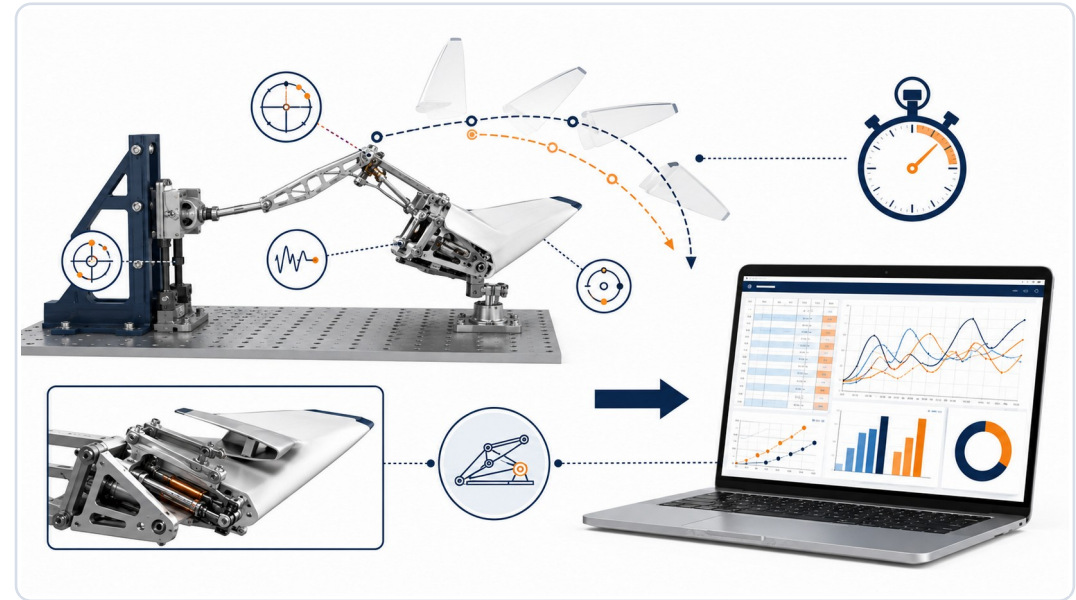


What you need to understand

How can you predict what might go wrong before testing a prototype?

Big ideas

- A failure mode is a specific way a design could fail.
- Risk depends on both likelihood and consequence.
- Good testing plans focus on the most important risks first.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 List failure modes

Think about breaking, slipping, tipping, binding, overheating, confusing use, or unsafe misuse.

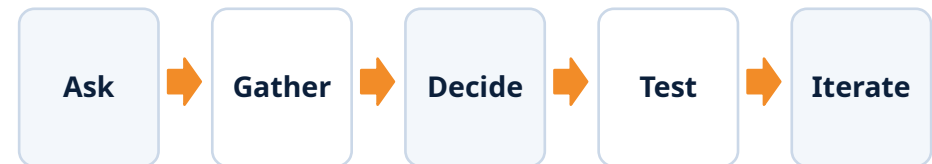
2 Rate risk

Rank the likelihood and severity of each failure.

3 Plan mitigation

Change the design, add instructions, strengthen a part, or improve the test.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Failure mode table

Evidence checklist

- Identify at least five possible failure modes.
- Rate risk for each.
- Choose two design changes or controls.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.9

Team Norms & Project Roles

Essential question

How do design teams organize work so every member contributes to the final solution?

What you will create

Team contract and role plan



What you need to understand

How do design teams organize work so every member contributes to the final solution?

Big ideas

- Team norms describe how the group communicates and makes decisions.
- Roles help divide responsibility without isolating team members.
- Strong teams document decisions and hold each other accountable.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Agree on norms

Set expectations for communication, attendance, shared work, and conflict.

2 Assign responsibilities

Use roles such as project manager, CAD lead, documentation lead, testing lead, and presentation lead.

3 Track decisions

Record what changed, why it changed, and who is responsible for next steps.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Team contract and role plan

Evidence checklist

- Write team norms.
- Assign roles and backups.
- Create a simple responsibility tracker.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.10

Project Management & Milestones

Essential question

How can a team break a large design project into manageable steps?

What you will create

Project timeline with milestones



What you need to understand

How can a team break a large design project into manageable steps?

Big ideas

- Milestones are checkpoints that show progress toward the final deliverable.
- A good plan includes dependencies: some tasks must happen before others.
- Project management is a design skill, not paperwork.



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Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Start with the deadline

Work backward from the final presentation and prototype due dates.

2 Build checkpoints

Include research, concept selection, CAD, prototype, test, iteration, and presentation milestones.

3 Adjust based on evidence

A plan should change when testing reveals new information.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Project timeline with milestones

Evidence checklist

- Create a milestone timeline.
- List task owners.
- Identify the next three actions for the team.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.11

Concept Generation for Client Problems

Essential question

How can teams create multiple solution ideas before choosing one direction?

What you will create

Individual and team concept sketches



How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Sketch individually first

Individual thinking prevents the group from settling too early.

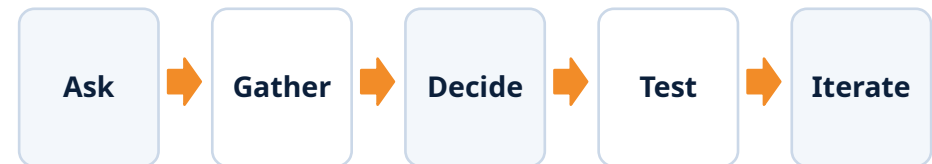
2 Share and combine

Look for useful features from multiple ideas.

3 Create concept families

Group ideas by strategy: support, protect, align, store, guide, move, or measure.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Individual and team concept sketches

Evidence checklist

- Create individual concept sketches.
- Share and annotate team ideas.
- Combine parts of multiple concepts into a team direction.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.12

Decision Matrices & Justification

Essential question

How can design teams choose a concept using evidence instead of preference?

What you will create

Decision matrix and recommendation



What you need to understand

How can design teams choose a concept using evidence instead of preference?

Big ideas

- A decision matrix compares concepts against weighted criteria.
- Scores should be based on evidence, not popularity.
- Justification explains why the selected idea best meets the problem.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Choose criteria carefully

Use requirements from the design brief, not random preferences.

2 Weight what matters most

Safety or usability may deserve more weight than appearance.

3 Explain the result

A matrix supports a decision, but the team must still justify tradeoffs.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Decision matrix and recommendation

Evidence checklist

- Build a decision matrix.
- Score at least three concepts.
- Write a short recommendation.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.13

Low-Fidelity Prototyping

Essential question

How can quick prototypes answer design questions before committing to a final build?

What you will create

Low-fidelity prototype and test notes



What you need to understand

How can quick prototypes answer design questions before committing to a final build?

Big ideas

- Low-fidelity prototypes are quick, rough, and useful.
- A prototype should answer a specific question.
- Early prototypes reduce risk before a team spends time on detailed CAD or fabrication.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Prototype one question

Focus on fit, reach, stability, layout, motion, or user interaction.

2 Use quick materials

Cardboard, paper, tape, scrap parts, or simple prints can reveal major issues.

3 Observe use

Watch how someone interacts with the prototype and record what happens.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Low-fidelity prototype and test notes

Evidence checklist

- Build or sketch a low-fidelity prototype.
- Write the question it tests.
- Record observations and improvement ideas.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.14

CAD Development & Prototype Planning

Essential question

How do teams move from concept to a manufacturable prototype?

What you will create

CAD plan and fabrication checklist



What you need to understand

How do teams move from concept to a manufacturable prototype?

Big ideas

- CAD should communicate design intent, not just shape.
- Prototype plans must consider dimensions, tolerances, materials, and assembly.
- Manufacturing decisions affect cost, time, quality, and reliability.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Build from requirements

Use criteria and constraints to decide features and dimensions.

2 Plan assemblies

Identify how parts connect, align, fasten, or move.

3 Check before fabrication

Review size, print orientation, material, tool access, and safety.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: CAD plan and fabrication checklist

Evidence checklist

- Create a CAD development plan.
- List parts to make or modify.
- Complete a fabrication readiness checklist.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.15

Testing Plans & Data Collection

Essential question

How can testing prove whether a design meets the requirements?

What you will create

Test plan and data table

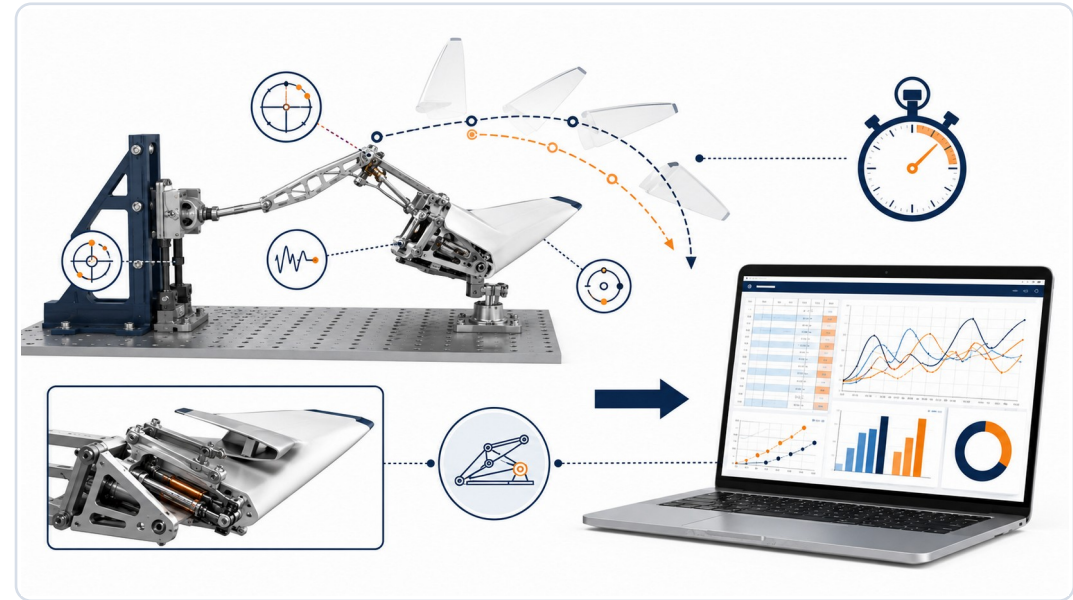


What you need to understand

How can testing prove whether a design meets the requirements?

Big ideas

- Testing must connect directly to success metrics.
- Good data is repeatable, organized, and tied to a clear procedure.
- A failed test is useful if it tells the team what to improve.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Write the procedure

Define setup, variables, number of trials, and what will be measured.

2 Collect clean data

Use tables, units, labels, and consistent measurement methods.

3 Compare to requirements

Decide whether results meet, exceed, or fall short of criteria.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Test plan and data table

Evidence checklist

- Write a test plan.
- Create a data table with units.
- Explain how results will be judged.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.16

Iteration & Final Design Decisions

Essential question

How should teams use feedback and test data to improve the design?

What you will create

Iteration log and final design update



What you need to understand

How should teams use feedback and test data to improve the design?

Big ideas

- Iteration is purposeful change based on evidence.
- A final design should reflect what the team learned from users and tests.
- Every major change should be documented and justified.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Review evidence

Use test results, user feedback, and failure modes to decide what must change.

2 Prioritize improvements

Fix high-risk, high-impact issues before cosmetic changes.

3 Document revisions

Show before/after changes and explain the reason for each.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Iteration log and final design update

Evidence checklist

- Create an iteration log.
- Identify the final design changes.
- Explain what evidence drove each change.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

What you need to understand

How can a design team clearly communicate the problem, evidence, solution, and impact?

Big ideas

- A strong presentation tells the story of the design process.
- Evidence matters more than describing every step.
- The audience needs to understand why the final solution is justified.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Start with the problem

Explain who needed help and why the problem mattered.

2 Show key evidence

Use research, requirements, testing, and iteration to support claims.

3 End with impact

Explain how the final design meets user needs and what could improve next.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Final presentation draft

Evidence checklist

- Draft final presentation slides.
- Add evidence visuals.
- Practice a concise explanation of the final design.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.

LESSON 5.18

Portfolio, Showcase & Reflection

Essential question

How can you use a finished project to show growth as an engineering designer?

What you will create

Final portfolio entry and reflection



What you need to understand

How can you use a finished project to show growth as an engineering designer?

Big ideas

- A portfolio shows process, not just the final object.
- Reflection helps you identify what you learned and what you would improve.
- Engineering evidence includes sketches, CAD, prototypes, tests, decisions, and communication.



Use visuals as evidence and inspiration, not as a final answer.



Strong designs start with a clear understanding of people, requirements, and evidence.

How to approach the work

Use a repeatable process so your design decisions are clear and defensible.

1 Select strong artifacts

Choose items that show research, design decisions, prototyping, testing, and iteration.

2 Explain your role

Describe what you contributed and how you helped the team improve.

3 Reflect honestly

Use specific evidence to show growth, challenges, and next steps.

Engineering habit



Evidence loops back into the next decision.

When you cannot explain why a design changed, you may be iterating randomly instead of engineering intentionally.

What your work should show

Use this lesson to produce: Final portfolio entry and reflection

Evidence checklist

- Create a final portfolio entry.
- Prepare for the showcase.
- Write an individual reflection on your engineering growth.

Quality check

Can another person understand what you decided, why it matters, and what evidence supports it?

Notebook / portfolio connection

Capture sketches, notes, tables, test results, photos, CAD screenshots, and reflections. Your final portfolio depends on evidence collected throughout the unit.



UNIT 5 WRAP-UP

Your final design story

A finished prototype is only part of the work. The complete engineering story includes the problem, users, requirements, decisions, tests, iteration, and impact.



Portfolio reminder

Show your process clearly enough that someone outside your team can understand and trust your decisions.

